

Hands-On AI: Build an Autonomous Agent with the Claude Agent SDK

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Prompts for the course

Video 2: Kickstart the deep research agent

We are building an agent using the Claude agent SDK in Python. We will do this in several steps, starting with a basic agent that responds to simple research and analysis queries. We will add memory and tools in future steps. So for now, let us just start by using the query approach for stateless interactions. No memory at this stage. Please add some health related research examples in the code to demonstrate how this research agent works. Let's say we are planning to do some research on "how our body digests protein" Ultrathink and read the documentation here before you proceed to implementation: <https://platform.claude.com/docs/en/agent-sdk/python>

Video 3: Add memory to create a deep research agent

Let's expand our agent's functionality and add conversational research & analysis with memory. Create some examples in the code relating to a multi-turn conversational research & analysis for illustration purposes. We want to move away from the stateless query approach to a more stateful and conversational approach. Refer to the code docs here: <https://platform.claude.com/docs/en/agent-sdk/python>

Video 4: Add web search and pdf downloader tools

Now expand our agent with some web search and document parsing tools. 1. Create a web search tool using tavily API that uses the query to search for relevant latest research papers, PDFs and articles from credible sources on that topic, go through tavily docs here: <https://docs.tavily.com/sdk/python/quick-start> . Use this tool for any web search related tasks. 2. Create a tool that downloads all the PDF from the "PDF" urls provided by the web search tool and store them in the local folder called "papers".

Video 5: Add looping to make an autonomous research agent

With tools now in place, let's make a fully autonomous research agent that works towards goals independently, without asking for questions. The user will feed the research topic from any field, and the agent works through with the tools available until the goal is achieved. The user needs to input enough information at step one in the example. Ultrathink the implementation for this agent and before writing code.

Video 6: Add a UI and turn the agent into an app

Create a production-ready web interface for the research agent using streamlit. It should be elegant, simple yet intuitive along with option to view PDFs.

An update to the agent behaviour: Every search from the user should create a dedicated folder for that query showcasing all the papers from that request.

This new folder structure should be visible in a left panel on the web interface. Ultrathink the implementation and then execute